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Aspire to Inspire the Digital World

Machine Learning and Data Mining

Ha Noi, 2026

- Course Title: Machine Learning and Data Mining
- Credits: 3 (45 hours)
- Learning Activities: 36 hours of lectures and 9 hours of exercises
- Prerequisites: Artificial Intelligence and Probability and Statistics
- Assessment Criteria:
  - Attendance: 5%
  - Self Learning on Edux: 5%
  - 10 Assignments (at least 5 on Edux): 20%
  - Midterm Exam: 20%
  - Final Exam: 50% (Project-based assignment)

Learner Obligations

- ✓ Attend at least 80% of the total class sessions for the course.
- ✓ Participate in group activities as specified by the course regulations.
- ✓ Independently study and investigate topics assigned by the instructor outside of class hours.
- ✓ Pre-class learning on Edux (edux.cmcu.edu.vn)
- ✓ Complete all assessments required for the course on EduX.
- ✓ Show respect to instructors and fellow learners.
- ✓ Comply with the academic integrity policies of the institution.

Learning Materials

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Primary Materials:

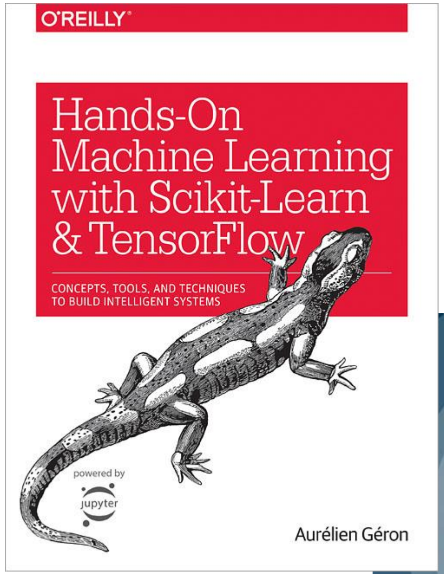
[1]. Zhang, A., Lipton, Z. C., Li, M., & Smola, A. J. (2023). Dive into Deep Learning. Cambridge University Press. https://d2l.ai

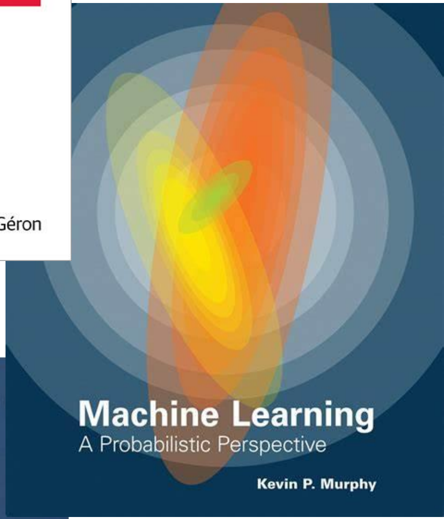
Reference Materials:

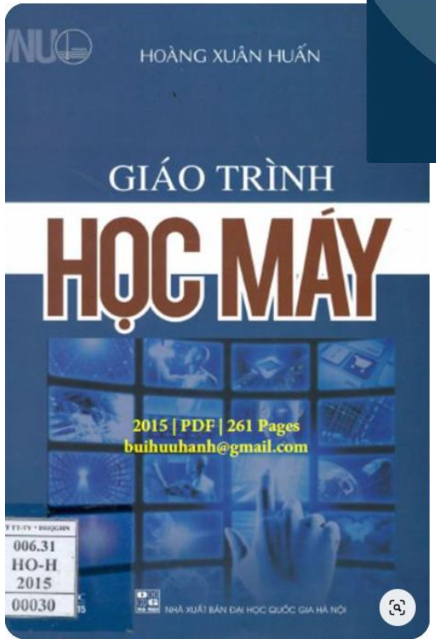
[2]. Kevin P. Murphy, Machine Learning: A Probabilistic Perspective, MIT Press, 2012.

[3]. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn and TensorFlow, O'Reilly, 2019.

[4]. Hoàng Xuân Huấn, Giáo trình học máy.







LESSON 1.
Machine Learning

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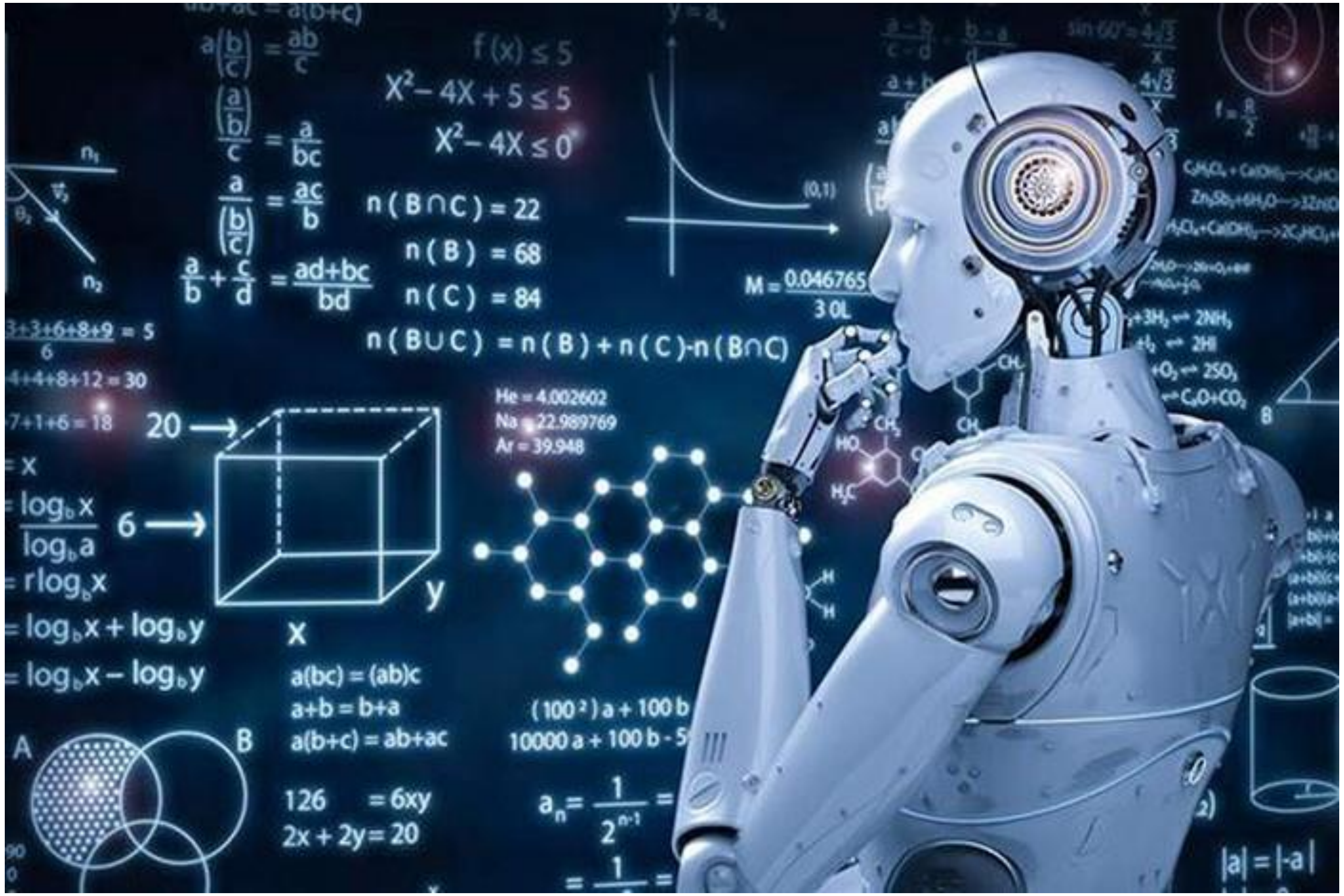
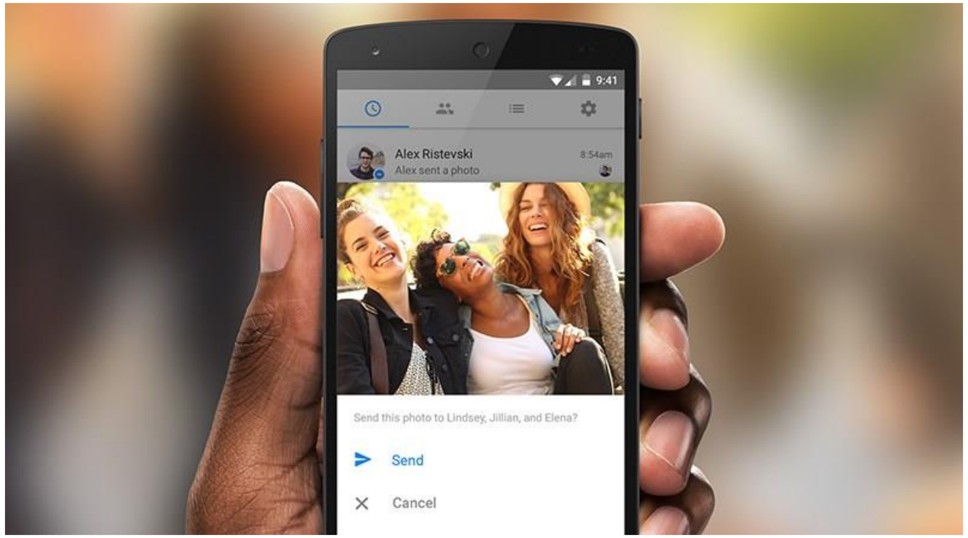
Machine Learning has emerged as one of the key manifestations of the Fourth Industrial Revolution



- Google self-driving cars.

- Face auto-tagging system in photos by Facebook.



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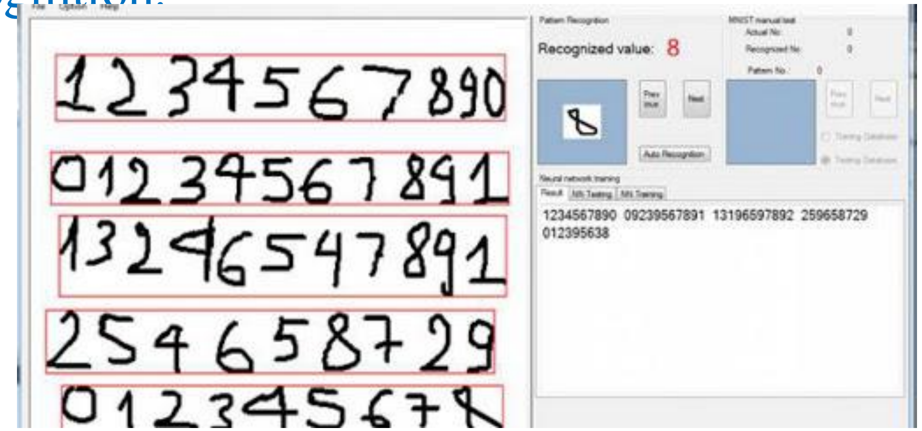
Some examples:

- Product recommendation system of Am

- Movie recommendation system of Netflix

- The AlphaGo Go-playing AI developed by Google DeepMind.

- Handwritten character recognition.



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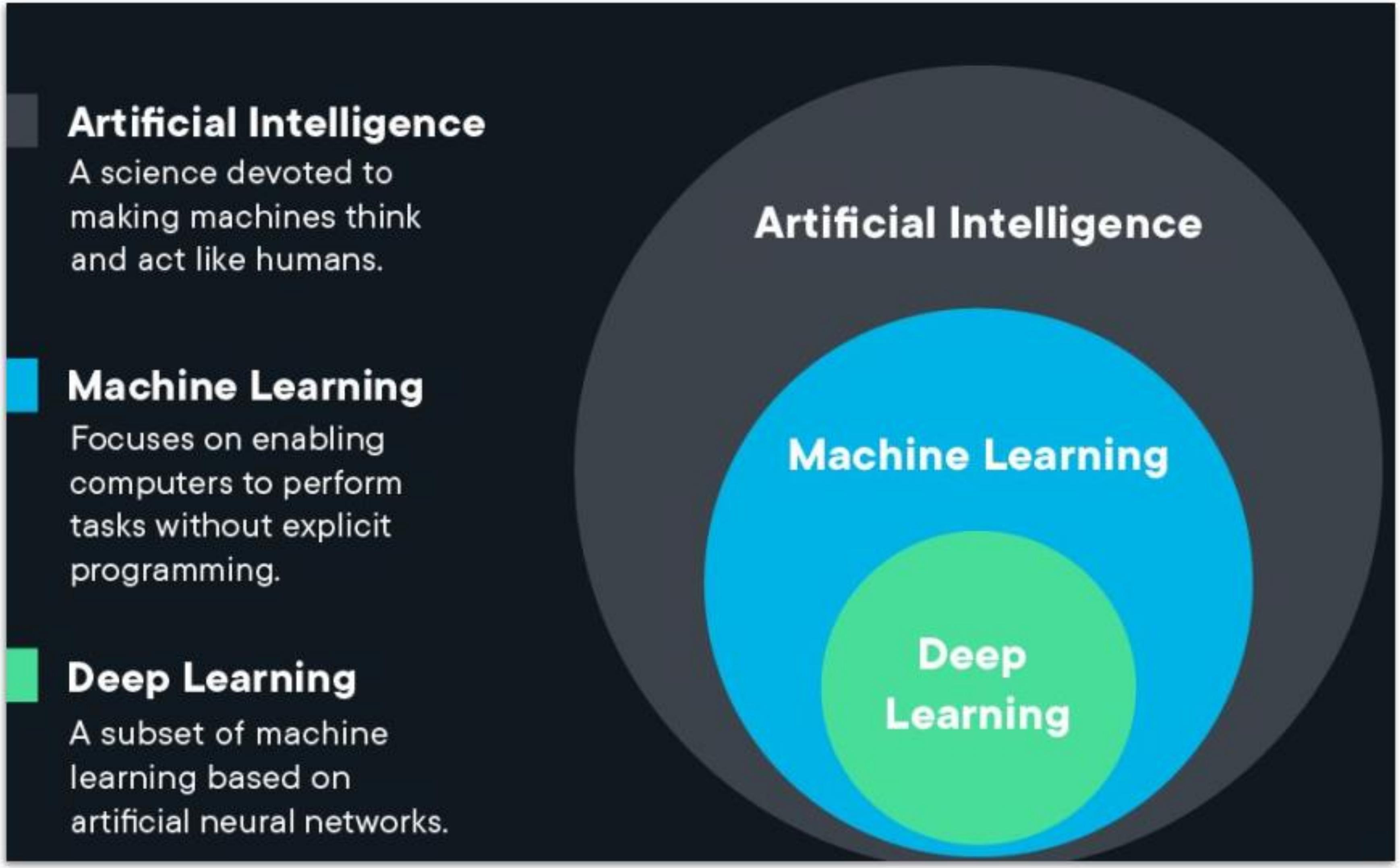
Machine Learning has the ability to learn automatically from input data without being explicitly programmed.

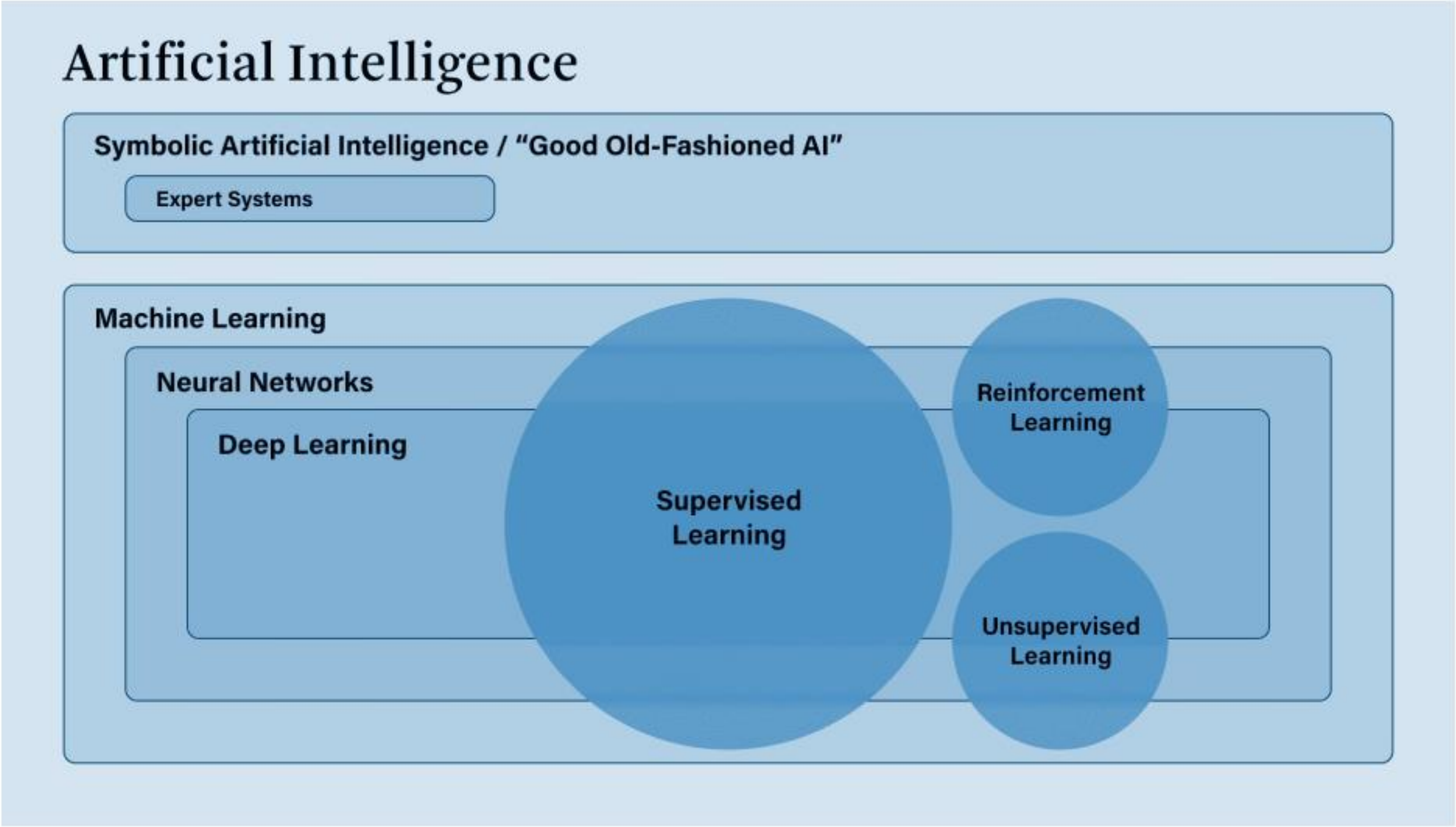


Classification of machine learning methods based on learning approaches:

- Supervised learning,
- Unsupervised learning,
- Semi-supervised lerning
- Reinforcement learning

Classification based on the functions of algorithms:

- Regression Algorithms
- Classification Algorithms
- Clustering Algorithms
- Bayesian Algorithms



Supervised Learning

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- Supervised learning is the most popular category of Machine Learning algorithms.
- Supervised learning is a method that predicts the output (outcome) of new data (new input) based on previously known (data, label) pairs.
- With a training example set.

Area (Data)	Number of bedrooms	Price (Label)
64	2	760
46	1	615
43	1	532
25	1	478

- Need to answer:

  How much will a house with x_1 m^2 , x_2 bedrooms cost?

Supervised Learning

- A set of input variables $X=\{x_1,x_2,\ldots,x_N\}$ and a corresponding set of labels $Y=\{y_1,y_2,\ldots,y_N\}$
- The known data pairs $(x_i,y_i) \in X\times Y$ are called the training data set.
- From the training data set, we need to build a function that maps each element from set X to a corresponding (approximate) element in set Y:

$$y_i \approx f(x_i),\ \ \forall i=1,2,\ldots,N$$

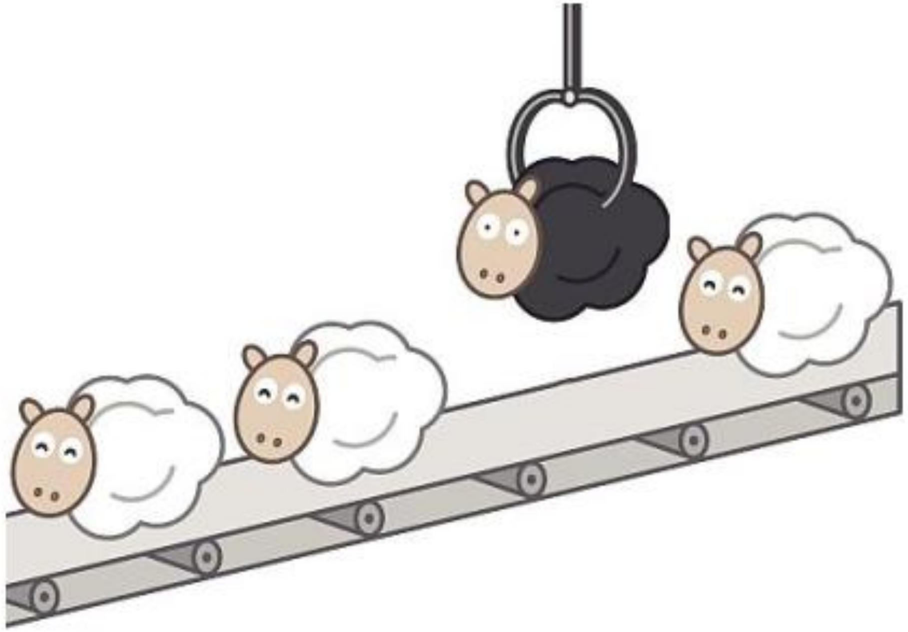
- The goal is to approximate the function f as accurately as possible so that when a new input x is given, we can predict its label $y=f(x)$

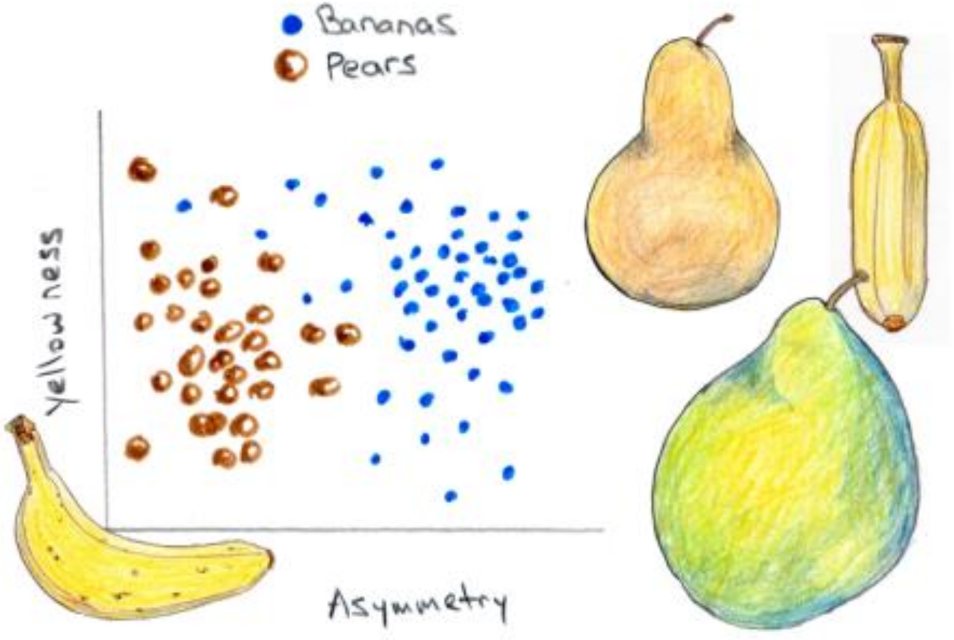
Classification: A problem is called a classification problem if the labels of the input data are divided into a finite number of categories.

Rec. ID	Age	Income	Student	Credit_Rating	Buy_Computer
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Medium	High	No	Fair	Yes
4	Old	Medium	No	Fair	Yes
5	Old	Low	Yes	Fair	Yes
6	Old	Low	Yes	Excellent	No
7	Medium	Low	Yes	Excellent	Yes
8	Young	Medium	No	Fair	No
9	Young	Low	Yes	Fair	Yes
10	Old	Medium	Yes	Fair	Yes
11	Young	Medium	Yes	Excellent	Yes
12	Medium	Medium	No	Excellent	Yes
13	Medium	High	Yes	Fair	Yes
14	Old	Medium	No	Excellent	No

Will a young student with an average income and a normal credit rating be classified into the “Yes” or “No” class?

Supervised Learning





Supervised Learning

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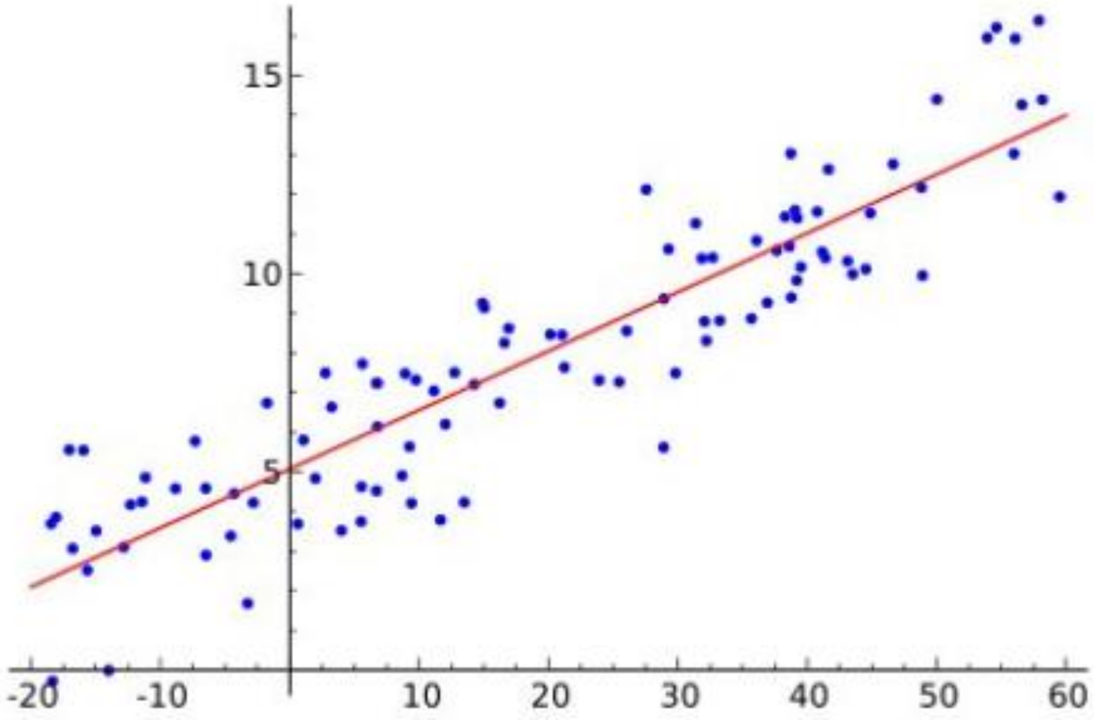
Regression: If the labels are not divided into categories but instead are specific real values.

Area	**Number of bedrooms.**	**Distance from Hoàn Kiếm Lake.**	**Price**
70	1	5 km	800 million VND.
90	2	5 km	1.2 billion VND.
120	3	15 km	1.1 billion VND.

Question:
How much will a house with an area of x_1 m^2 ; x_2 bedrooms, and located x_3 km from Hoàn Kiếm Lake cost?

Supervised Learning

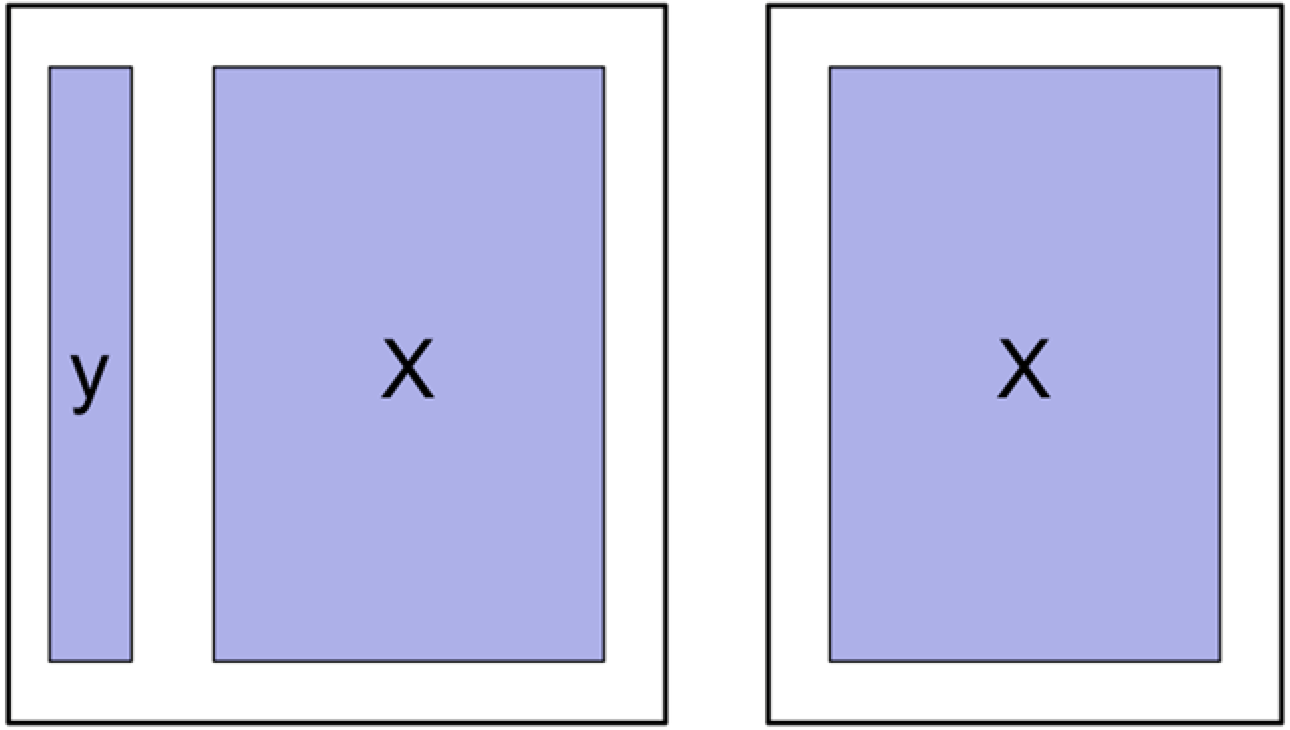


Unsupervised Learning

- Unsupervised learning is a learning approach where we only have input data X without knowing the labels Y.
- Unsupervised learning algorithms rely on the structure of the data to perform certain tasks, such as:
  - Clustering
  - Dimension reduction

Food Item	Spiciness	Sweetness
Milk tea	1	5
Vanilla ice cream	1	4
Spicy beef noodle soup	5	1
Spicy fried chicken	4	1

Unsupervised Learning



Some mathematical notations.

Italic letters represent scalar values.

$$x_1, N, y, k$$

Bold lowercase letters represent vectors.

$$\mathbf{y}, \mathbf{x}_1$$

Bold uppercase letters represent matrices.

$$\mathbf{X}, \mathbf{Y}, \mathbf{W}$$

Some mathematical notations.

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$\mathbf{x} = [x_1, x_2, \ldots, x_n]$ is a row vector.

$\mathbf{x} = [x_1; x_2; \ldots; x_n]$ is a column vector.

$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n]$ is a matrix whose x_j are column vectors.

$\mathbf{X} = [\mathbf{x}_1; \mathbf{x}_2; \ldots; \mathbf{x}_m]$ is a matrix whose x_j are row vectors.

x_{ij} is the element at row i, column j.

$\mathbb{R}^n$ denotes the set of column vectors with n elements.

$\mathbb{R}^{m \times n}$ denotes the set of matrices with m rows and n columns.

$\mathbf{w}_i$ is the i-th column vector of matrix W.

The L_2-norm of a vector.

The length of a vector $\mathrm{x} \in \mathbb{R}^n$ is its L_2-norm.

$$\|\mathbf{x}\|_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

$$\|\mathbf{x}\|_2^2 = \mathbf{x}^T\mathbf{x}.$$